

MT 504: Powder Metallurgical Processing of Materials

Course Code	MT 504
Course Name	Powder Metallurgical Processing of Materials
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech/PG/Ph.D.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction:** Historical background, important steps in the powder metallurgy (P/M) process, advantages and limitations of the powder metallurgy process, and applications. [6 Lectures]
- **Powder Production:** Production methods like physical, chemical, and mechanical methods; Single fluid atomization like rotating electrode atomization, roller atomization, and rotating disc atomization; Two fluid atomizations like gas atomization, water atomization, oil atomization, etc. Reduction methods, carbonyl process, hydride-dehydride process, electrolytic method. Mechanical milling, fluid energy grinding, machining. Production of ceramic powders. [8 Lectures]
- **Powder Characteristics:** Sampling-chemical composition, particle shape and size analysis, Surface area, packing and low characteristics, Porosity and density, compressibility, and strength properties. Blending and mixing of metal powders. [8 Lectures]
- **Powder Compaction and Sintering:** Compaction of powders, pressure less and pressure compaction techniques-single action and double action compaction, cold compaction, powder rolling, continuous compaction, explosive compaction, hot temperature compaction, uniaxial hot pressing, hot extrusion, spark sintering, hot isostatic pressing, injection moulding, sintering types, theory of sintering, process variables, effects of sintering, sintering atmospheres, and metallographic technique for sintered products. [8 Lectures]
- **Post-sintering operations:** sizing, coining, repressing, and resintering; impregnation; infiltration; Heat treatment, steam treatment, machining, joining, plating; and other coatings. [6 Lectures].
- **Application of powder Metallurgy & products:** Self-lubricating bearings, magnetic materials, tungsten carbide tool bits, bearing materials, dispersion-strengthening materials for high temperature applications, and manufacture of diamond-based cutting tools. [6 Lectures]

Laboratory Modules

1. NA

Text Books

1. Masuda H., **Powder Technology Handbook**, Taylor & Francis, 2006.
2. Angelo, P.C., and R. Subramanian, **Powder metallurgy science, technology, and applications**, Prentice Hall Publishers, 2008.

References

1. German R.M., **A to Z of Powder Metallurgy**, Elsevier 2005.
2. Sands R.L. and Shakespeare C.R., **Powder Metallurgy Practice and Applications**, Newness Publication 1970.
3. **Powder Metal Technologies and Applications**, Metals Handbook, Vol. 7, 9th 1989 edition, ASM
4. Hirschhorn J.S., **Introduction to Powder Metallurgy**, APMI 1975
5. Upadhyaya G.S., **Powder Metallurgy Technology**, Cambridge Press 1996